Discriminative Marginalized Least-Squares Regression for Hyperspectral Image Classification

Yuxiang Zhang, *Student Member, IEEE*, Wei Li, *Senior Member, IEEE*, Heng-Chao Li, *Senior Member, IEEE*, Ran Tao, *Senior Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— Least-squares regression (LSR)-based classifiers are effective in multiclassification tasks. However, most existing methods use limited projections, resulting in loss of much discriminant information; furthermore, they focus only on exactly fitting samples to target matrix while ignoring overfitting issue. To solve these drawbacks, discriminative marginalized LSR (DMLSR) is proposed to learn a more discriminative projection matrix with consideration of class separability and data-reconstruction ability simultaneously. In the proposed framework, an intraclass compactness graph is employed to avoid the overfitting problem and enhance class separability, and a data-reconstruction constraint is imposed to preserve discriminant information on limited projections. Experimental results on several hyperspectral data sets demonstrate that the proposed method significantly outperforms some state-of-the-art classifiers.**

***Index Terms*— Hyperspectral image, least-squares regression (LSR), manifold regularization, pattern classification, regression-based classification.**

I. Introduction

HYPERSPECTRAL imagery (HSI) have wideband range and are used to identify different vegetation and materials [1]–[6]. Based on rich spectral signatures, numerous classification techniques have been investigated [7], [8] and used in many remote sensing applications, such as urban mapping and environment pollution monitoring [9]–[13].

Manuscript received June 25, 2019; revised September 19, 2019 and October 15, 2019; accepted October 16, 2019. Date of publication October 31, 2019; date of current version April 22, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant NSFC-91638201 and Grant 61571033, in part by the Beijing Natural Science Foundation under Grant 4172043, in part by the Beijing Nova Program under Grant Z171100001117050, and in part by the Fundamental Research Funds for the Central Universities under Grant 3052019116 *(Corresponding author: Wei Li.)*

Y. Zhang is with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing 100029, China (e-mail: zyx829625@163.com).

W. Li and R. Tao are with the School of Information and Electronics, Beijing Institute Technology, Beijing 100081, China (e-mail: liwei089@ieee.org; rantao@bit.edu.cn).

H.-C. Li is with the School of Information Science and Technology, Southwest Jiaotong University, Chengdu 610031, China (e-mail: lihengchao_78@163.com).

Q. Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Mississippi State, MS 39762 USA (e-mail: du@ece.msstate.edu).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2019.2949082

Recently, representation-based classifiers have been widely studied [14]–[17], including sparse representation (SR), collaborative representation (CR), and low-rank representation (LRR). SR-based classifier (SRC) was initially proposed for face recognition [18]. The core idea of the SRC is to assume that each testing sample can be approximated by sparse linear combination of an overcompleted dictionary. Although SRC has considerable performance, it suffers from high computational complexity, and many scholars have suggested that the role of sparsity in recognition is limited [19]. Thus, efficient CR-based classifier (CRC) has been proposed, which has a closed-form solution and less complexity. A testing sample is represented by the collaborative linear combination of all training samples in CRC. In addition, the emergence of LRR-based method is due to the fact that SRC only focuses on finding SR of a single sample and ignores global structure of data. LRR can identify potential correlations between samples and retain data membership.

However, current representation-based methods are not flexible and adaptive enough in machine learning tasks, such as classification or regression, because these methods are weak in distinguishing and heavy in calculation. They focus on the validity and accuracy of data representation, but the best representation does not necessarily imply the best separability or discrimination [20]. Moreover, these methods do not transform original data features into discriminant feature representations, resulting in the learned data representation being not optimal for classification. Besides, SR- and LRR-based methods cannot meet the requirements of real-time processing due to the limitation of their computational complexity.

Least-squares regression (LSR) using class labels has received much attention in pattern recognition field, such as face recognition [21], feature representation [22], and feature selection [23]. Linear regression (LR), one of the simplest regression, is different from the aforementioned representation-based methods, which can closely link the original features of data to the target output to learn more discriminant data representation. In multiclassification tasks, it uses class labels to define a binary matrix as the regression target and learns a projection matrix that can transform samples into their respective label spaces. However, the discrete binary (i.e., zero-one) label matrix in the standard LR is so rigid that it is not suitable for classification tasks and learning

0196-2892 © 2019 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

transformed samples. To solve the problem, one strategy is to employ other regression target. For instance, a ε-dragging technique was developed for relaxing the regression target [24], and a retargeted LSR (ReLSR) was proposed [25], which learned a soft target matrix from data by enforcing marginalized constraint without using the ε-dragging technique.

Although these methods reflect promising classification performance, target overfitting issue is inevitable. From this perspective, a discriminative LSR (DLSR) [24] forced the regression targets of true and false classes moving along the opposite directions by using the ε-dragging technique, which made the transformed samples overfitting the targets to obtain large interclass distance. Furthermore, an interclass sparsity-based DLSR (ICS-DLSR) was proposed [26], where an interclass sparsity constraint was integrated into standard LSR to maintain the row sparsity consistency structure and to mine relationships among the samples, further improving class separability. A regularized label relaxation (RLR) was developed [27], in which the method not only relaxed the target matrix but also introduced manifold learning to avoid overfitting and preserved intraclass relationship in the transformed space. A marginally structured representation learning (MSRL) method was developed [28], where an adaptive probabilistic graph structure was constructed by capturing probabilistic connectivity between the samples to improve the compactness of transformed samples.

All the abovementioned LSR-based classifiers have achieved satisfying classification accuracy on face recognition; however, they encounter two limitations that greatly degrade the performance when directly applied to hyperspectral data. On the one hand, most of them pursue the largest margins while assuming that the training samples should be accurately projected into the targets matrix (such as a binary matrix), but ignoring the overfitting problem [27], that is, when the same ground objects have different spectral reflectances (intraclass variation), the classifier overly fits a few pixels that deviate from normal samples and loses the main features. The effective way of avoiding overfitting is to ensure the compactness among the intraclass samples after projection. On the other hand, they only can project original samples into c-dimensional subspace because the label matrix is used as the regression target (c represents the number of classes, which is small compared with the original band of HSI). In other words, the transformed samples fail to preserve the main energy of hyperspectral data in limited projections, resulting in a loss of discriminant information, which is not conducive to the classification performance.

In order to overcome the two limitations mentioned earlier, both class separability and data-reconstruction ability are considered to learn robust latent subspace and more discriminative projection. Discriminative marginalized LSR (DMLSR) is developed. The proposed DMLSR classifier is different from the traditional ReLSR, which is given in the following.

1) As a mathematical form of intraclass samples structure, intraclass compactness graph is integrated into ReLSR to capture the potential geometric relationships of intraclass samples and guide the projection matrix learning.
2) Considering the abovementioned limitation, important information will be lost in the subspace of finite dimensions, and a data-reconstruction constraint is imposed to preserve the main features of transformed samples.

The main contributions of this article are summarized as follows.

1) The intraclass compactness graph is imposed on the projection matrix to avoid the overfitting issue and further enhance class separability.
2) A data-reconstruction constraint is designed to learn the transformed samples with main discriminant information and further assist compactness graph to improve the ability to prevent targets overfitting, which makes the proposed method more robust than other LSR-based methods. As far as our knowledge, there exist very few LSR-based classifiers for hyperspectral image analysis. The proposed DMLSR avoids target overfitting, preserves discriminant information simultaneously, and further learns a robust latent subspace and a more discriminative projection matrix, resulting in better classification performance.

The rest of this article is organized as follows. Section II describes the basic regression-based classifiers. In Section III, the detailed description and optimal solution of the proposed DMLSR are provided. In Section IV, comparison performance of the proposed method with several existing LSR-based and representation-based classifiers is present. Finally, some remarkable conclusions are drawn in Section V.

II. Regression-Based Classification

Due to effectiveness and simplicity, LSR-based classifiers have been extensively studied [29], [30]. The traditional LSR classifier is designed to minimize the regression errors and learn linear projection matrix, which is expected to accurately fit the original labeled training samples $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{n} \in \mathbb{R}^d$ (d is spectral dimension) with the labels $l \in \{1, 2, \ldots, c\}$ to binary label matrix and effectively obtain the largest margins. The model can be formulated as

$$\min_{\mathbf{Q}} \|\mathbf{A} - \mathbf{X}^T\mathbf{Q}\|_F^2 + \lambda\|\mathbf{Q}\|_F^2 \tag{1}$$

where λ is a positive regularization parameter, $\mathbf{A} \in \mathbb{R}^{n\times c}$ is a label matrix composed of 0 and 1 values with the jth element of the ith row being equal to 1 and other elements being equal to 0 when the ith sample $\mathbf{x}_i$ belongs to the jth class, and $\mathbf{Q} \in \mathbb{R}^{d\times c}$ is the learned projection matrix. The abovementioned function has a closed-form solution as

$$\mathbf{Q} = (\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I})^{-1}\mathbf{X}\mathbf{A}. \tag{2}$$

The label of $\mathbf{y} \in \mathbb{R}^{d\times 1}$ is predicted by $\max_{l=1,\ldots,c} (\mathbf{y}^T\mathbf{Q})_l$ [i.e., $\text{class}(\mathbf{y}) = \arg\max_{l=1,\ldots,c} (\mathbf{y}^T\mathbf{Q})_l$], where $(\mathbf{y}^T\mathbf{Q})_l$ denotes the lth element of vector $\mathbf{y}^T\mathbf{Q}$.

LSR model often learns a projection matrix by establishing projection relationship between the labeled training samples and the strict zero-one label matrix. However, a zero-one label matrix is too strict to learn an effective projection matrix for fitting class labels, further greatly reducing the regression

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

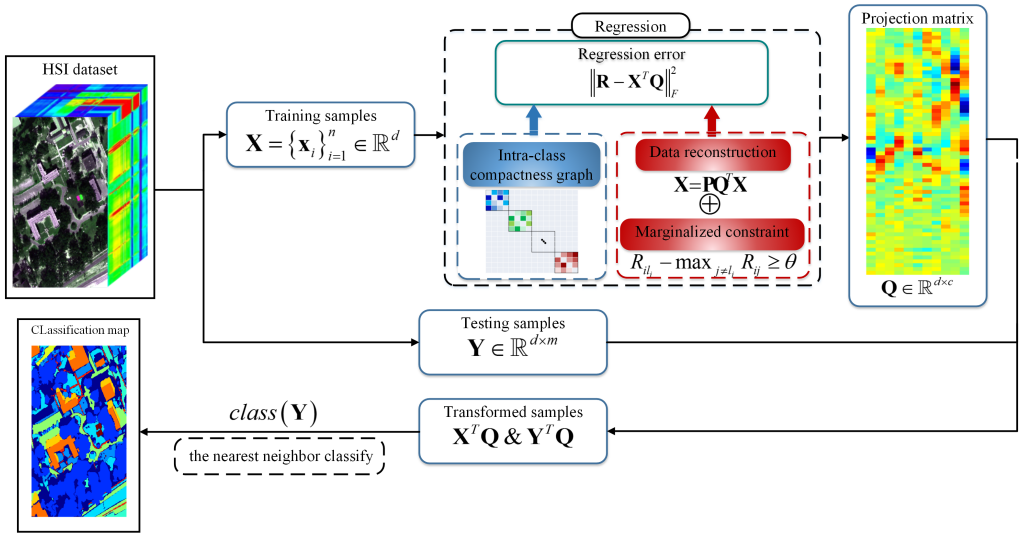


Fig. 1. Flowchart of the proposed DMLSR, including the marginalized constraint, data reconstruction, and compactness graph.

accuracy of the model and limiting the flexibility [25]. To solve this problem, ReLSR has been developed to seek an adaptively regression target, where a soft label matrix is learned from data in place of the zero-one label matrix that is used as regression targets. ReLSR learns jointly the projection matrix $\mathbf{Q}$ and the regression target matrix $\mathbf{R}$ by optimizing the following problem:

$$\min_{\mathbf{Q},\mathbf{R},\mathbf{b}} \|\mathbf{R}-\mathbf{X}^T\mathbf{Q}-\mathbf{e}^T\mathbf{b}\|_F^2+\lambda\|\mathbf{Q}\|_F^2 \quad \text{s.t.} \quad \{R_{il_i}-\max_{j\neq l_i}R_{ij}\geq 1\}_{i=1}^{n} \tag{3}$$

where $\mathbf{b}\in\mathbb{R}^{1\times c}$ is an offset vector, $\mathbf{e}=[1,1,\ldots,1]\in\mathbb{R}^{1\times n}$ is a column vector, and l_i denotes the index of the true class for the sample $\mathbf{x}_i$. The soft regression target matrix is expressed as $\{\mathbf{R}|\{R_{il_i}-\max_{j\neq l_i}R_{ij}\geq 1\}_{i=1}^{n}\}$. Obviously, the marginalized constraint increases the interclass boundary of the transformed samples in ReLSR, which makes it possible to learn a discriminant projection matrix.

Some extensions, including DLSR and ReLSR, only focus on fitting the soft label matrix with the training samples accurately while ignoring the fact that a simple fitting may break internal structure of data. To address the drawback, an interclass sparsity constraint has been integrated into the traditional LSR [26], i.e., ICS-DLSR, which aims at preserving a row sparsity consistency structure in each class

$$\min_{\mathbf{Q},\mathbf{E}} \frac{1}{2}\|\mathbf{A}+\mathbf{E}-\mathbf{Q}\mathbf{X}\|_F^2+\frac{\lambda_1}{2}\|\mathbf{Q}\|_F^2+\lambda_2\sum_{l=1}^{c}\|\mathbf{Q}\mathbf{X}_l\|_{2,1}+\lambda_3\|\mathbf{E}\|_{2,1} \tag{4}$$

where λ_1, λ_2, and λ_3 are the regularization parameters, $\sum_{l=1}^{c}\|\mathbf{Q}\mathbf{X}_l\|_{2,1}$ is the interclass sparsity constraint, and $\mathbf{E}$ denotes the sparsity error term to relax the zero-one label matrix. This constraint brings a promising class sparsity, and the row sparsity consistency structure improves the class separability, which is beneficial to the multiclassification.

In addition, RLR [27] was developed to maintain the data manifold structure and prevent target overfitting. In RLR,

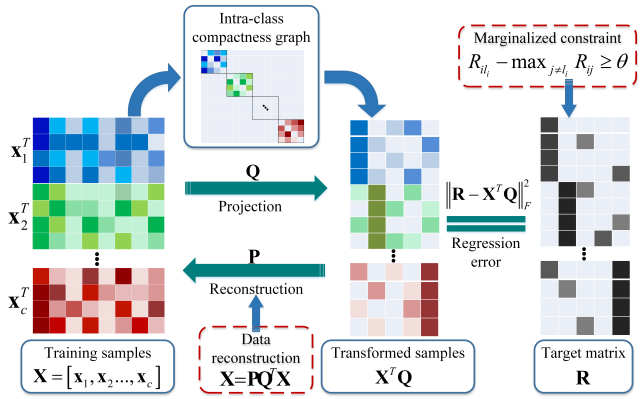


Fig. 2. Schematic of obtaining soft regression target matrix process.

a compactness graph based on manifold learning is established to preserve intrinsic geometric structure of training samples, and the nonnegative dragging matrix of DLSR is also introduced

$$\min_{\mathbf{Q},\mathbf{M}} \|\mathbf{X}^T\mathbf{Q}-(\mathbf{A}+\mathbf{B}\odot\mathbf{M})\|_F^2+\lambda\mathrm{tr}(\mathbf{Q}^T\mathbf{X}\mathbf{L}\mathbf{X}^T\mathbf{Q}) \quad \text{s.t.} \quad \mathbf{M}\geq 0 \tag{5}$$

where $\mathbf{M}$ is a nonnegative dragging matrix, $\odot$ is a Hadamard product operator, $\mathbf{L}$ is a Laplacian matrix, and $\mathbf{B}$ is a luxury matrix and defined as

$$\mathbf{B}_{i,j}=\begin{cases}+1, & \text{if} \quad \mathbf{L}_{i,j}=1\\ -1, & \text{otherwise.}\end{cases} \tag{6}$$

Although extensive experiments have proved the classification performance of RLR, it cannot effectively solve the overfitting problem in a low-dimensional data [27].

III. Proposed DMLSR Framework

Fig. 1 shows the flowchart of the proposed DMLSR, where training samples are fit to the target matrix with the constraint of data reconstruction and compactness graph, so as to learn the discriminatively transformed samples.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

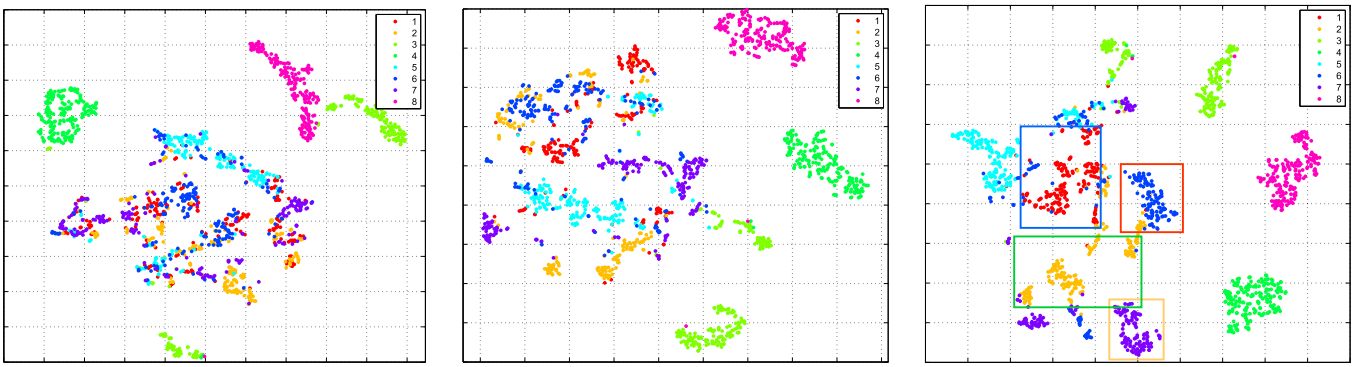


Fig. 3. 2-D visualization of the Indian Pines data using the selected samples (eight classes in total). (a) Original samples. (b) Transformed samples by RLR. (c) Transformed samples by DMLSR.

A. Algorithm Formulation

The proposed method inherits from the original ReLSR, which is to learn soft regression target matrix directly from data by imposing a marginalized constraint. Due to marginalized constraint force, the distance between the correct and incorrect classes, implementation of this constraint makes the learned transformed samples with larger interclass margins and enhances the flexibility of regression model. However, the ReLSR may excessively fit a few pixels that deviate from normal samples due to the target relaxation when intraclass variations exist in real HSI data. The proposed approach addresses the problem by imposing an intraclass compactness graph, which can preserve intraclass relationship of the original samples and reduce the intraclass distances between the abnormal and normal samples in the transformed space, further reinforcing the learned transformed samples separability. The problem can be formulated as

$$\begin{aligned} &\min_{\mathbf{Q},\mathbf{R}} \|\mathbf{R}-\mathbf{X}^T\mathbf{Q}\|_F^2+\lambda_1\|\mathbf{Q}\|_F^2+\lambda_2\mathcal{T} \\ &\text{s.t. } \{R_{il_i}-\max_{j\neq l_i} R_{ij}\geq\theta\}_{i=1}^n \end{aligned} \tag{7}$$

where $\mathbf{R}=[\mathbf{R}_1,\mathbf{R}_2,\ldots,\mathbf{R}_i,\ldots,\mathbf{R}_n]^T\in\mathbb{R}^{n\times c}$ is the learned soft regression target matrix and θ is a constant. It has been experimentally verified that the difference of the two marginal values (corresponding to the correct class that the ith sample $\mathbf{x}_i$ belongs to and the other incorrect class that the ith sample $\mathbf{x}_i$ does not belong to) of each $\mathbf{R}_i$ is optimally set to 0.1, i.e., $\theta=0.1$. λ_1 and λ_2 are two regularization parameters, and $\mathcal{T}$ is the manifold regularization term corresponding to intraclass compactness graph $\mathbf{W}$. The edge weight $\mathbf{W}_{ij}$ of the graph $\mathbf{W}$ is calculated from the Gaussian function $\mathbf{W}_{ij}=\exp(-(\|\mathbf{x}_i-\mathbf{x}_j\|_2^2/t_it_j))$ if $\mathbf{x}_i$ and $\mathbf{x}_j$ from the same class, 0 otherwise, where $t_i=\|\mathbf{x}_i-\mathbf{x}_i^{(k_{nn})}\|$ denotes the local scaling of data samples in the neighborhood of $\mathbf{x}_i$ [31], and reflects the similarity between a central pixel and neighboring pixels. When the local similarity is higher, the difference value is lower, resulting in a larger weight of the central pixel $\mathbf{x}_i$. In addition, $\mathbf{x}_i^{(k_{nn})}$ is the k_{nn}-nearest neighbors of $\mathbf{x}_i$, and the best performance of $k_{nn}=4$ for all experiments was verified experimentally. The graph-preserving criterion is defined as

$$\min_{\mathbf{Q}}\sum_{i\neq j}\|\mathbf{Q}^T\mathbf{x}_i-\mathbf{Q}^T\mathbf{x}_j\|^2\mathbf{W}_{i,j}. \tag{8}$$

For the sample $\mathbf{x}_i$ and sample $\mathbf{x}_j$ from the same class with great similarity, the corresponding edge weight $\mathbf{W}_{ij}$ is also large, and the transformed samples $\mathbf{Q}^T\mathbf{x}_i$ and $\mathbf{Q}^T\mathbf{x}_j$ with smaller distances are obtained by minimizing (8). Therefore, the compact samples can be learned from the similarity preservation property of (8). The manifold regularization term $\mathcal{T}$ is obtained by converting (8) into the form of a trace

$$\begin{aligned} \min_{\mathbf{Q}}\sum_{i\neq j}\|\mathbf{Q}^T\mathbf{x}_i-\mathbf{Q}^T\mathbf{x}_j\|^2\mathbf{W}_{i,j}&=\min_{\mathbf{Q}}Tr(\mathbf{Q}^T\mathbf{X}\mathbf{L}\mathbf{X}^T\mathbf{Q}) \\ \Rightarrow\ \mathcal{T}&=Tr(\mathbf{Q}^T\mathbf{X}\mathbf{L}\mathbf{X}^T\mathbf{Q}) \end{aligned} \tag{9}$$

where $\mathbf{L}$ is a Laplacian matrix, and $\mathbf{L}=\mathbf{D}-\mathbf{W}$, where $\mathbf{D}$ a diagonal matrix with the ith diagonal element being defined as $\mathbf{D}_{ii}=\sum_{j=1}^n\mathbf{W}_{i,j}$.

Although the model in (7) introduces manifold regularization term to solve target overfitting problem, a fixed number of projections may result in the loss of many discriminant information contained in a multiband. To solve these problems, a data-reconstruction constraint is further designed into the abovementioned model to enhance the reconstruction ability of transformed samples

$$\begin{aligned} &\min_{\mathbf{Q},\mathbf{R},\mathbf{P}}\|\mathbf{R}-\mathbf{X}^T\mathbf{Q}\|_F^2+\lambda_1\|\mathbf{Q}\|_F^2+\lambda_2Tr(\mathbf{Q}^T\mathbf{X}\mathbf{L}\mathbf{X}^T\mathbf{Q}) \\ &\text{s.t. } \{R_{il_i}-\max_{j\neq l_i}R_{ij}\geq\theta\}_{i=1}^n,\quad \mathbf{X}=\mathbf{P}\mathbf{Q}^T\mathbf{X},\ \mathbf{P}^T\mathbf{P}=\mathbf{I} \end{aligned} \tag{10}$$

where $\mathbf{P}\in\mathbb{R}^{d\times c}$ is an orthogonal reconstruction matrix. The method of data reconstruction through two different matrices rather than a single matrix $\mathbf{Q}$ has been proven to be effective [32], in which matrix $\mathbf{Q}$ can better fit the target matrix and $\mathbf{P}$ has more freedom to ensure that the transformed samples keep the main energy in the fixed c-dimensional subspace. Through the combination of data-reconstruction term and manifold regularization term, less information loss can be guaranteed while maintaining the manifold structure of data,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

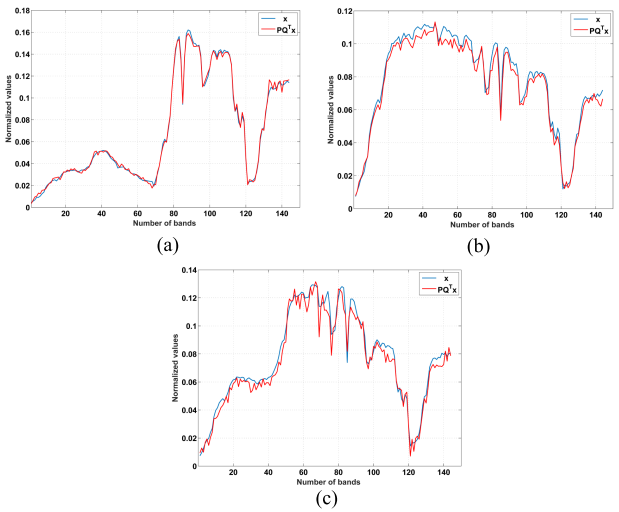


Fig. 4. Spectral curve of three pixels in the Houston data showing the reconstruction capability of the proposed algorithm. (a) Pixel from the fourth class. (b) Pixel from the ninth class. (c) Pixel from the 15th class.

TABLE I
SIMILARITY MEASUREMENT OF SELECTED SAMPLES USING THE HOUSTON DATA BEFORE AND AFTER RECONSTRUCTION

Evaluation index	RMSE	Cross-correlation	SAM
Score	0.0068	0.9847	0.3968

which can better prevent overfitting in a finite projections space and thus obtain a more robust subspace. After iteratively optimizing variables and obtaining $\mathbf{Q}$, the testing sample $\mathbf{y}$ is classified via performing the nearest neighbor method based on $\mathbf{y}^T\mathbf{Q}$.

B. Algorithm Optimization

The classic alternating direction method of multipliers (ADMM) is employed to optimize iteratively the proposed DMLSR [33]. First, (10) can be solved by minimizing the corresponding augmented Lagrange multiplier formulation

$$\begin{aligned}&\min_{\mathbf{Q},\mathbf{R},\mathbf{P}} \|\mathbf{R}-\mathbf{X}^T\mathbf{Q}\|_F^2+\lambda_1\|\mathbf{Q}\|_F^2\\&\quad+\lambda_2 Tr(\mathbf{Q}^T\mathbf{G}_1\mathbf{Q})+\mu\left\|\mathbf{X}-\mathbf{P}\mathbf{Q}^T\mathbf{X}+\frac{\gamma}{\mu}\right\|_F^2\\&\text{s.t. } \{R_{il_i}-\max_{j\neq l_i}R_{ij}\geq\theta\}_{i=1}^n,\quad \mathbf{P}^T\mathbf{P}=\mathbf{I}\end{aligned}\tag{11}$$

where $\mathbf{G}_1=\mathbf{X}\mathbf{L}\mathbf{X}^T$, and γ and μ are the Lagrange multiplier and the penalty parameter, respectively. The detailed optimization steps are as follows.

Updating $\mathbf{Q}$*:* Fixing other variables and update $\mathbf{Q}$ by solving the following problem:

$$\begin{aligned}\mathbf{Q}=\arg\min_{\mathbf{Q}}\ &\|\mathbf{R}-\mathbf{X}^T\mathbf{Q}\|_2^2+\lambda_1\|\mathbf{Q}\|_F^2\\&+\lambda_2 Tr(\mathbf{Q}^T\mathbf{G}_1\mathbf{Q})+\mu\left\|\mathbf{X}-\mathbf{P}\mathbf{Q}^T\mathbf{X}+\frac{\gamma}{\mu}\right\|_F^2\end{aligned}\tag{12}$$

whose solution can be inferred by setting the first partial derivative with respect to $\mathbf{Q}$ and making it to zero.

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE INDIAN PINES DATA

Class		Number of Samples	
No.	Name	Train	Test
1	Corn-no-till	187	1247
2	Corn-min-till	187	647
3	Soybean-no-till	187	310
4	Soybean-min-till	187	302
5	Soybean-clean-till	187	781
6	Grass/Pasture	187	2281
7	Hay-windowed	187	427
8	Woods	187	1107
Total		1496	7102

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE MISSISSIPPI DATA

Class		Number of Samples	
No.	Name	Train	Test
1	trees	150	23096
2	grass pure	150	4120
3	grass ground surface	150	6732
4	dirt and sand	150	1676
5	road materials	150	6537
6	water	150	316
7	shadow building	150	2083
8	buildings	150	6090
9	sidewalk	150	1235
10	yellowcurb	150	33
11	cloth panels	150	119
Total		2706	51423

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON DATA

Class		Number of Samples	
No.	Name	Train	Test
1	Health grass	198	1053
2	Stressed grass	190	1064
3	Synthetic grass	192	505
4	Tress	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking lot 1	192	1041
13	Parking lot 2	184	285
14	Tennis court	181	247
15	Running track	187	473
Total		2832	12197

After calculation, an expression for $\mathbf{Q}$ is obtained

$$\mathbf{Q}=[\lambda_1\mathbf{I}+(1+\mu)\mathbf{X}\mathbf{X}^T+\lambda_2\mathbf{G}_1]^{-1}(\mu\mathbf{X}\mathbf{G}_2^T\mathbf{P}+\mathbf{X}\mathbf{R})\tag{13}$$

where $\mathbf{G}_2=\mathbf{X}+\gamma/\mu$ and $\mathbf{I}$ is an identity matrix of appropriate size.

Updating $\mathbf{R}$: Fixing $\mathbf{Q}$ and $\mathbf{P}$, the optimization of $\mathbf{R}$ is described as

$$\min_{\mathbf{R}}\|\mathbf{R}-\mathbf{X}^T\mathbf{Q}\|_F^2\quad\text{s.t. } R_{il_i}-\max_{j\neq l_i}R_{ij}\geq\theta\tag{14}$$

where the constant θ is set to 0.1. The problem in (14) is a convex-constrained quadratic programming one [34], which is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

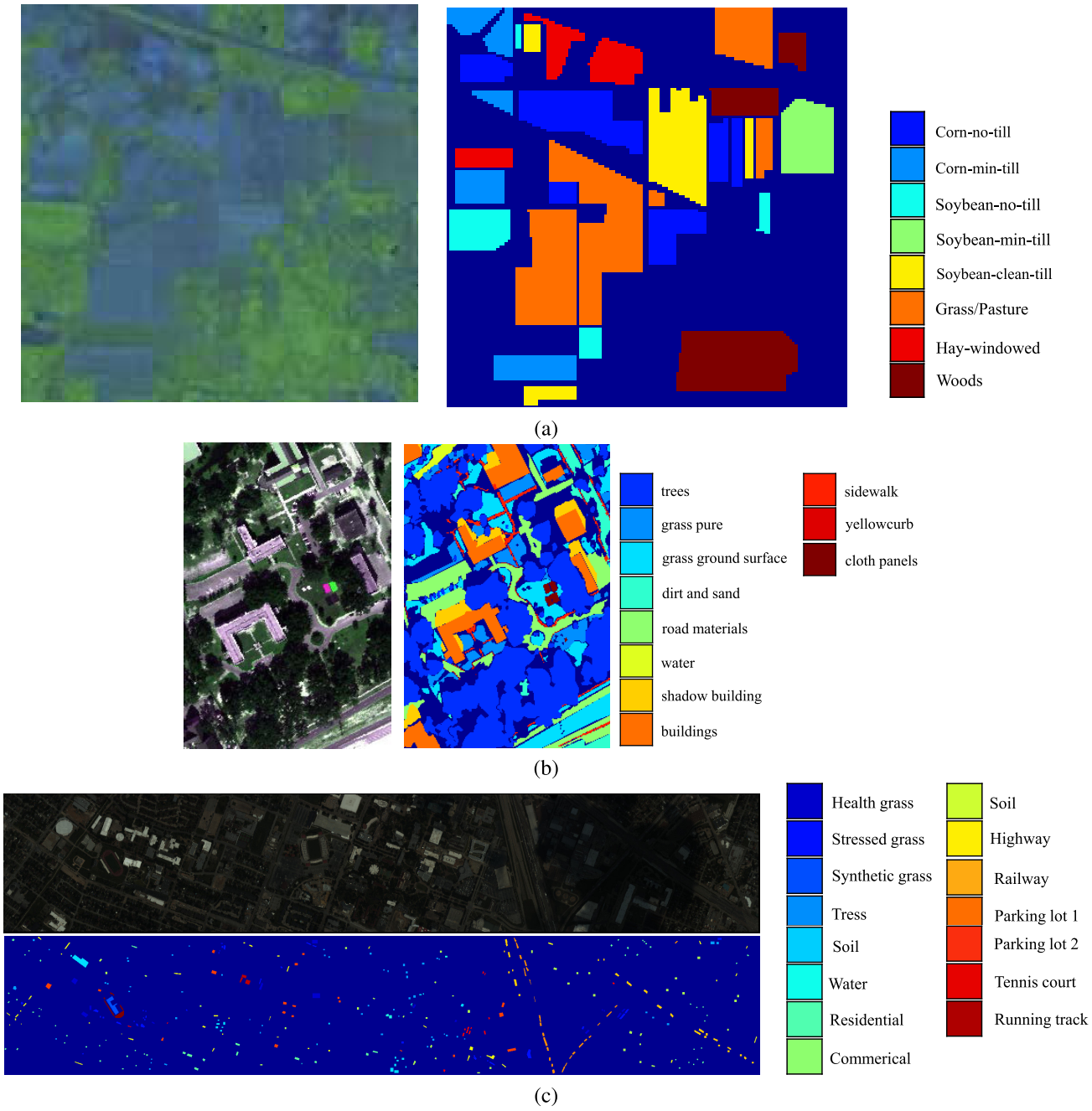


Fig. 5. False-color image and ground-truth mapof the three experimental data. (a) Indian Pines. (b) Mississippi. (c) Houston.

decomposed into n independent subproblems for estimating $\mathbf{R}$. Define $\mathbf{H} = \mathbf{X}^T\mathbf{Q}$, $\mathbf{H} = [\mathbf{H}_1, \mathbf{H}_2, \ldots, \mathbf{H}_n]^T \in \mathbb{R}^{n\times c}$. Given the ith sample is from the kth class, the ith row of $\mathbf{R}$ and $\mathbf{H}$ is marked as $\mathbf{R}_i = [r_1, r_2, \ldots, r_c] \in \mathbb{R}^c$ and $\mathbf{H}_i = [h_1, h_2, \ldots, h_c] \in \mathbb{R}^c$, respectively; the value of the kth element R_{ik} of $\mathbf{R}_i$ is denoted as r_k; and the ith subproblem in (14) becomes

$$\min_{\mathbf{R}_i} \frac{1}{2}\|\mathbf{R}_i - \mathbf{H}_i\|_F^2 = \sum_{j=1}^{c}(r_j - h_j)^2 \quad \text{s.t. } r_k - \max_{j\neq k} r_j \geq 0.1. \tag{15}$$

The complete optimization process of $\mathbf{R}$ is described in [25].

Updating $\mathbf{P}$*:* Similarly, by fixing $\mathbf{Q}$ and $\mathbf{R}$, the optimization problem of $\mathbf{P}$ can be written as

$$\min_{\mathbf{P}} \|\mathbf{G}_2 - \mathbf{P}\mathbf{Q}^T\mathbf{X}\|_F^2 \quad \text{s.t. } \mathbf{P}^T\mathbf{P} = \mathbf{I}. \tag{16}$$

The problem in (16) is an orthogonal procrustes one, which can be solved by the singular value decomposition (SVD) as $SVD(\mathbf{G}_2\mathbf{X}^T\mathbf{Q}) = \mathbf{U}\mathbf{S}\mathbf{V}^T$, and the solution of $\mathbf{P}$ is $\mathbf{P} = \mathbf{U}\mathbf{V}^T$.

Updating γ *and* μ*:* γ and μ are updated as

$$\gamma = \gamma + \mu(\mathbf{X} - \mathbf{P}\mathbf{Q}^T\mathbf{X}) \tag{17}$$

$$\mu = \min(\rho\mu, \mu_{\max}) \tag{18}$$

where ρ and $\mu_{\max}$ are constants. The procedure of the proposed DMLSR is summarized in Algorithm 1.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

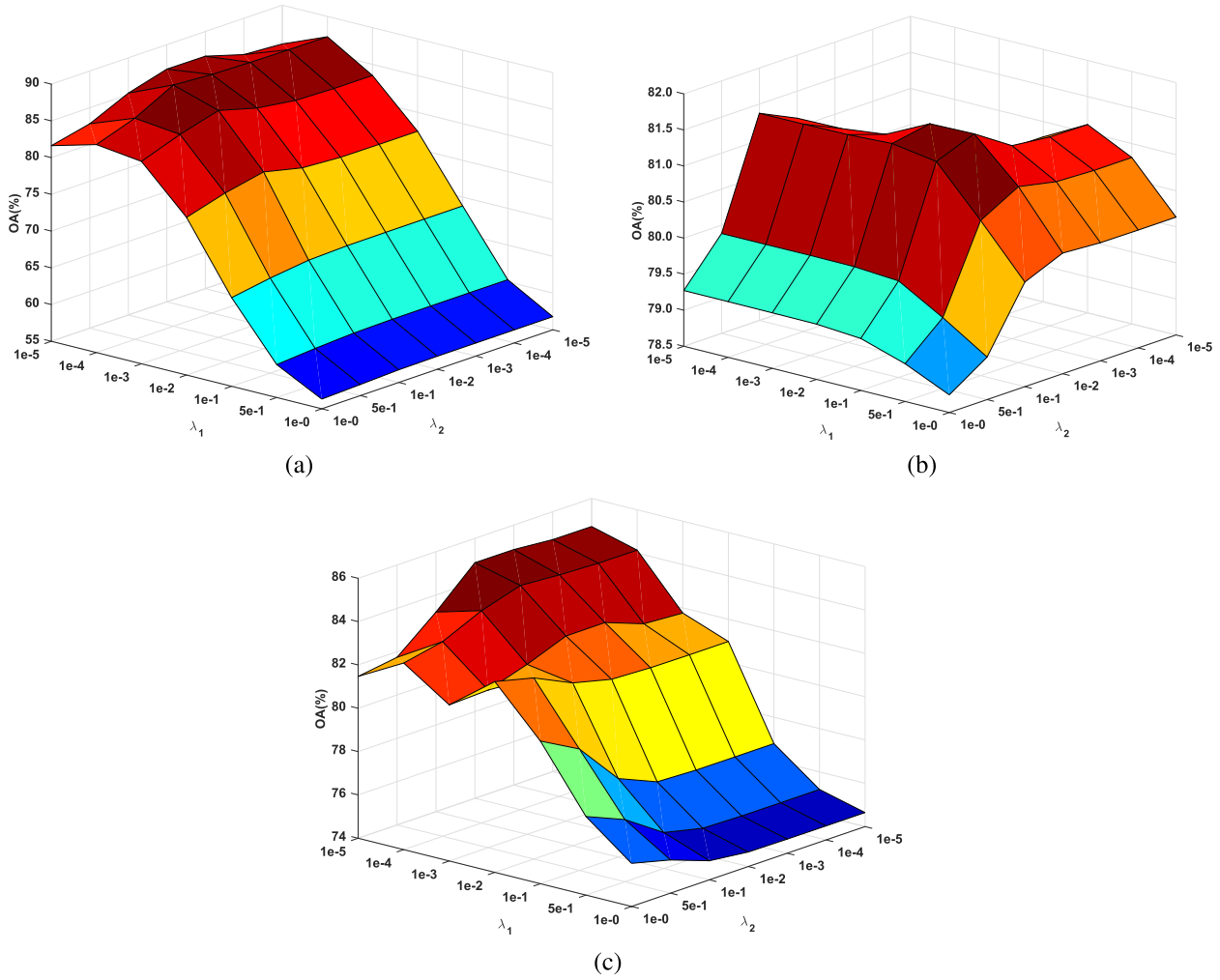


Fig. 6. Parameter tuning of λ_1 and λ_2 for the proposed DMLSR using the three experimental data. (a) Indian Pines. (b) Mississippi. (c) Houston.

Algorithm 1 Proposed DMLSR Classification Framework

Input: Training data $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^n$, label matrix $\mathbf{L}$, and regularization parameters λ_1, λ_2.
Initialization:Matrix $\mathbf{Q}$ with random values, $\mathbf{F} = \mathbf{QX}$, $\mathbf{T} = \mathbf{L}$, $\mu = 1e-8$, $\gamma = 0$ and $iter = 0$.
Step 1: Calculate $\mathbf{W}$ and $\mathbf{L}$
while not converged **or** max iter **do**
  Step 2: Update the projection matrix $\mathbf{Q}$ according to (13)
  Step 3: Update the target matrix $\mathbf{R}$ row-by-row
  Step 3: Update the reconstruction matrix $\mathbf{P}$ by solving (16)
  Step 5: Update the multiplier γ and the penalty parameter μ according to (17) and (18)
end while
Output: $\mathbf{Q}$, $\mathbf{P}$.

C. Analysis of the Proposed Method

For real HSI analysis, pixels from the same class may be characterized by different spectra, which causes the LSR-based classifier easily overfitting. Fig. 3(a) shows such phenomena, which is the 2-D visualization of the original samples obtained from the Indian Pines data and each color represents a class. The sample distribution of some classes is not concentrated or even heavily dispersed, and there are samples mixed together between these classes, such as the first class (i.e., *Corn-no-till*), the second class (i.e., *Corn-min-till*), and the sixth class (i.e., *grass/pasture*). The loose distribution of samples from the same class indicates that there is a high variation between the samples from the same class, i.e., intraclass variation. Fig. 3(b) and (c) shows the transformed samples by RLR and DMLSR, respectively, and it is obvious that much better class separability is achieved through DMLSR than the original samples. Therefore, the compactness of intraclass samples is significantly improved than RLR, which can better solve the overfitting problem.

To demonstrate the quantitative effect of the proposed DMLSR, spectral curves of three training samples from different classes using the Houston data are shown in Fig. 4, where the blue curve is the original sample (i.e., $\mathbf{x}$) and the red curve (i.e., $\mathbf{PQ}^T\mathbf{x}$) is the reconstructed one. The root-mean-square error (RMSE), cross correlation, and spectral angle measures (SAMs) are employed to evaluate the similarity between the original samples and the reconstructed samples, and the corresponding scores are listed in Table I. These results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

TABLE V

CA AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE INDIAN PINES DATA

Class	Classification algorithms											
	SVM	SRC	CRC	OTVCA	SSLRCA	DLSR	ReLSR	ICS-DLSR	MSRL	RLSL	RLR	DMLSR
1	82.28	80.43	79.63	78.97	80.50	81.40	81.72	82.44	84.36	83.88	84.28	85.72±1.10
2	89.34	75.43	81.45	88.34	92.69	83.93	84.08	84.23	88.41	84.23	82.69	89.98±1.89
3	97.42	97.42	95.81	96.62	95.95	97.42	97.10	97.74	97.42	97.10	98.06	97.64±0.86
4	100	100	100	100	98.63	100	100	100	100	100	100	100±0
5	86.94	68.25	74.65	94.78	78.22	82.07	84.25	82.84	84.12	81.18	84.25	91.41±1.31
6	75.49	72.82	73.48	82.28	83.82	71.24	74.31	71.64	74.44	71.42	70.50	77.53±2.16
7	94.38	87.82	92.04	83.50	96.80	89.23	90.63	91.57	94.15	88.99	92.04	95.57±0.39
8	99.19	99.73	99.28	99.81	95.64	98.37	98.74	99.28	99.28	99.19	99.19	99.33±0.03
OA (%)	86.03	81.22	82.65	87.76	87.11	83.05	84.47	83.77	85.67	83.57	83.77	**88.03±0.23**
KC (κ)	83.12	77.22	78.95	85.13	84.32	79.55	81.21	80.40	82.66	80.14	80.41	**85.27±0.32**

TABLE VI

CA AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE MISSISSIPPI DATA

Class	Classification algorithms											
	SVM	SRC	CRC	OTVCA	SSLRCA	DLSR	ReLSR	ICS-DLSR	MSRL	RLSL	RLR	DMLSR
1	83.85	77.65	86.89	84.54	81.52	81.60	86.21	81.69	82.75	82.84	82.72	83.15±2.89
2	75.22	44.44	60.15	83.62	81.55	77.38	77.96	76.02	76.04	75.49	75.53	77.89±4.57
3	66.67	39.25	69.09	63.29	66.98	66.83	68.52	67.93	66.71	67.59	66.79	67.35±3.15
4	81.09	75.42	89.20	78.46	84.25	85.62	83.17	84.01	84.79	84.79	85.02	86.96±2.82
5	83.34	42.33	43.71	83.88	85.07	78.48	66.83	81.05	80.57	83.63	81.17	85.15±1.69
6	96.20	75.00	87.97	96.52	97.78	96.20	91.14	97.47	96.52	98.42	97.15	95.83±1.58
7	80.56	60.06	76.81	86.70	87.76	78.59	73.02	78.44	80.41	80.84	78.83	82.61±2.60
8	79.10	75.39	89.23	85.81	86.35	82.23	75.52	84.81	82.92	82.81	82.09	83.26±1.56
9	72.06	41.46	59.35	53.68	63.00	68.18	62.83	69.47	69.07	70.53	67.85	71.81±4.75
10	100	96.97	100	66.67	66.67	100	96.97	100	100	100	100	96.97±2.85
11	99.16	97.48	99.16	96.64	94.96	97.48	96.64	97.48	97.48	98.32	97.48	98.60±2.17
OA (%)	79.95	63.76	76.38	81.03	80.67	78.86	78.47	79.54	79.67	80.21	79.52	**81.17±0.27**
KC (κ)	74.08	54.06	69.25	**75.55**	75.27	72.82	72.05	73.68	73.78	74.46	73.59	75.37±0.34

TABLE VII

CA AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE HOUSTON DATA

Class	Classification algorithms											
	SVM	SRC	CRC	OTVCA	SSLRCA	DLSR	ReLSR	ICS-DLSR	MSRL	RLSL	RLR	DMLSR
1	81.86	80.44	80.53	82.24	82.15	81.86	81.10	81.20	82.43	81.48	81.39	82.91
2	82.61	81.30	83.55	82.05	78.85	82.61	82.71	81.30	82.42	82.05	82.99	95.77
3	99.80	99.60	99.01	99.60	98.22	100	100	100	100	100	100	100
4	92.05	87.59	96.88	91.86	86.93	98.11	95.45	97.25	97.82	96.78	96.69	98.39
5	98.39	91.95	81.82	97.73	99.05	99.34	99.53	98.96	99.43	99.24	99.62	99.34
6	94.41	93.71	86.01	94.41	93.01	99.30	99.30	99.30	97.90	95.10	99.30	99.30
7	76.87	80.13	56.53	79.29	85.35	82.93	79.66	77.24	84.70	75.75	84.89	83.86
8	43.02	67.14	69.42	36.47	49.10	63.25	73.41	69.14	63.63	70.18	61.54	72.84
9	79.04	69.97	42.59	66.38	81.21	73.56	69.12	71.01	69.50	73.37	73.84	71.10
10	58.01	52.22	31.37	48.07	42.28	53.19	72.30	70.66	76.06	75.48	55.98	80.50
11	81.59	54.84	26.94	70.68	57.50	73.53	69.54	65.28	74.86	83.11	74.29	72.11
12	72.91	84.15	37.66	56.96	67.63	70.32	62.25	65.51	76.75	72.43	69.74	71.18
13	71.23	67.72	89.12	58.25	49.47	72.98	72.28	72.28	74.04	71.93	75.09	72.28
14	99.60	99.60	98.38	99.19	99.60	99.60	99.60	99.60	99.19	99.60	100	99.60
15	97.67	96.41	94.29	98.73	99.58	98.73	98.73	98.73	98.73	97.89	98.52	98.73
OA (%)	79.00	77.41	65.42	74.03	75.39	80.23	80.74	80.09	82.69	82.77	80.48	**84.45**
KC (κ)	77.41	75.58	62.76	72.12	73.52	78.54	79.09	78.38	81.21	81.30	78.84	**83.11**

validate the data-reconstruction capability, which ensures that the transformed samples $\mathbf{Q}^T\mathbf{X}$ hold the primary data energy.

IV. EXPERIMENTAL RESULTS

Three real hyperspectral data sets (i.e., the Indian Pines data, the Mississippi data, and the Houston data) are conducted to validate the effectiveness of the proposed method. Several classic and state-of-the-art classifiers are considered as comparison, including support vector machine (SVM), SRC [15], CRC [19], sparse and smooth low-rank analysis (SSLRA) [35], orthogonal total variation component analysis (OTVCA) [36], DLSR [24], DLSR [24], ReLSR [25], ICS-DLSR [26], MSRL [28], robust latent subspace learning (RLSL) [32], and RLR [27], where SVM is a general nonlinear version, and the kernel adopts radial basis function (RBF). For a fair comparison, SSLRA, OTVCA, and other LSR-based algorithms adopt the nearest neighbor classifier. The class-specific accuracy (CA), the overall accuracy (OA), and the Kappa coefficient (KC) are employed to assess the classification performance.

A. Experimental Data

1) Indian Pines Data Set: The data were selected from the scene at the Indiana Pines proving ground in Northwest Indiana and captured using NASA's airborne Airborne Visible/Infrared Imaging Spectroradiometer (AVIRIS) sensor in June 1992. The image scene includes crops and natural vegetation, with 145 × 145 pixels, a spatial resolution of 20 m,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

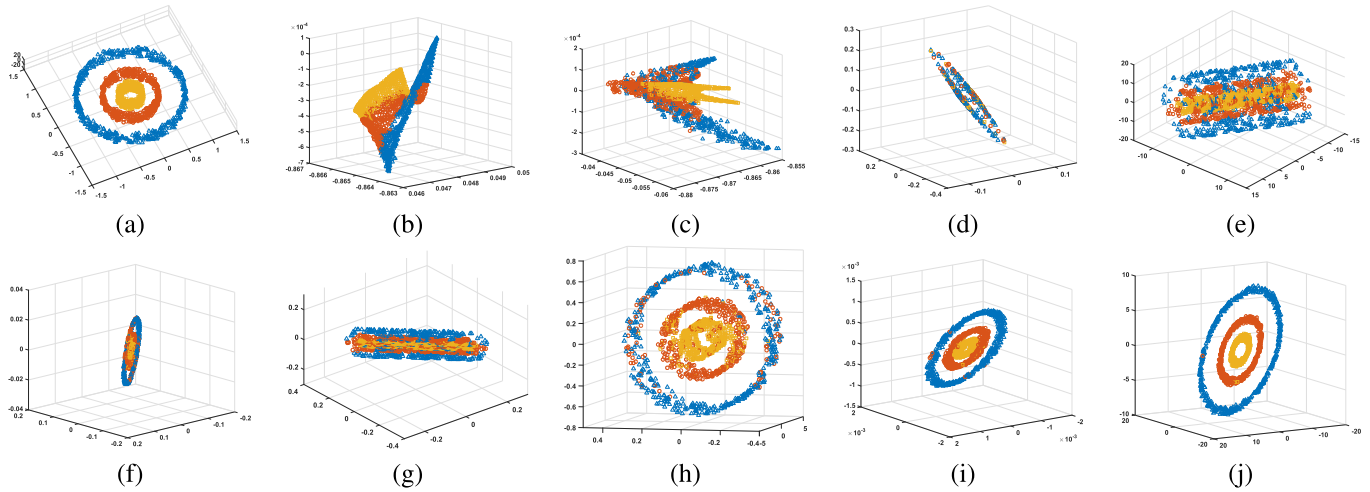


Fig. 7. Visualization and classification results for the three-ring data obtained with different methods including (a) three-ring data, (b) SSLRCA (97.93%), (c) OTVCA (78.74%), (d) DLSR (56.33%), (e) ReLSR (65.33%), (f) ICS-DLSR (77.00%), (g) MSRL (65.13%), (h) RLSL (85.33%), (i) RLR (91.13%), and (j) DMLSR (99.67%).

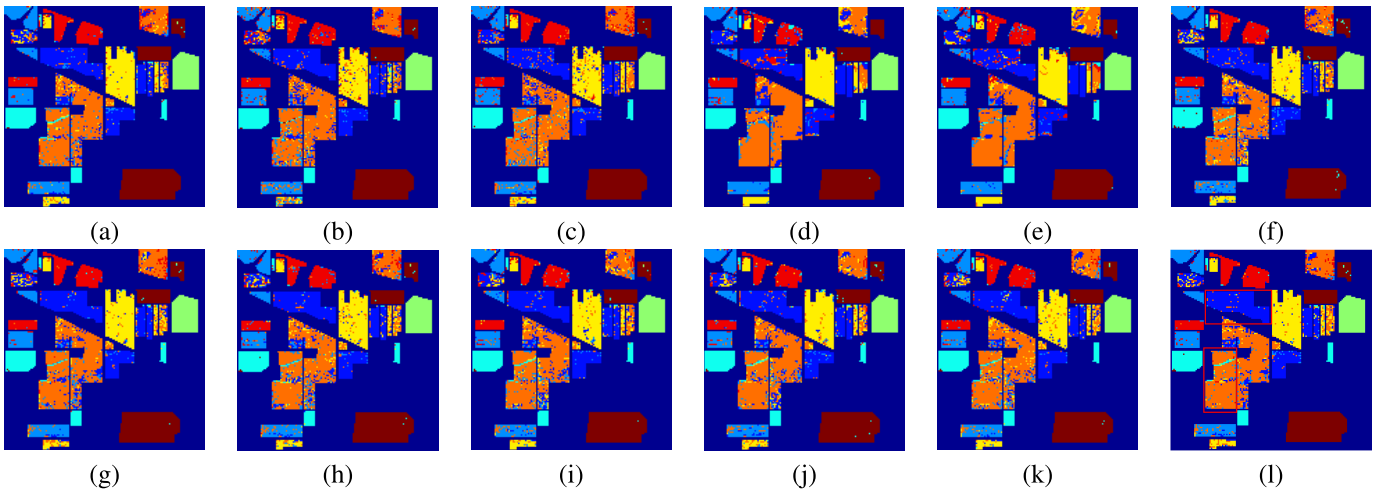


Fig. 8. Visualization and classification maps for the Indian Pine data obtained with different methods including (a) SVM (86.03%), (b) SRC (81.22%), (c) CRC (82.65%), (d) OTVCA (87.76%), (e) SSLRCA (87.11%), (f) DLSR (83.05%), (g) ReLSR (84.47%), (h) ICS-DLSR (83.77%), (i) MSRL (85.67%), (j) RLSL (83.57%), (k) RLR (83.77%), and (l) DMLSR (88.03%).

a spectral range from visible light to middle infrared light with 220 bands, and a spectral wavelength range of 0.4–2.45 μm. Here, 202 bands of water band removal were used. In the original truth map, there were 16 different land cover types, and we select eight classes to analyze more training samples from the perspective of statistics [37]. The data set is divided into training and testing pixels and the class-specific numbers are listed in Table II. Additionally, a false-color representation and ground-truth map are shown in Fig. 5(a).

2) Mississippi Data Set: The data were collected in November 2010 over the University of Southern Mississippi Gulf Park Campus, including 72 bands. Due to noise, the first four and last four bands were removed, resulting in a new hyperspectral image of 64 bands. The size of the cropped image is 325 $\times$ 220. The ground truth contains 11 classes [38], and the numbers of training and testing samples are listed in Table III. In addition, a false-color representation and ground truth map are shown in Fig. 5(b).

Houston Data Set: The data were captured by the airborne sensor on the University of Houston campus and surrounding areas in 2012. It consists of hyperspectral and Lidar images and was introduced in the 2013 GRSS Data Fusion Competition. The image scene contains vegetation, water bodies, and buildings with 349 $\times$ 1905 pixels and has a wavelength range of 0.38–1.05 μm and 144 spectral bands. In addition, the spatial resolution is 2.5 m, the ground truth includes 15 classes [39], and the numbers of training and testing samples are listed in Table IV. A false-color representation and ground truth map are shown in Fig. 5(c).

B. Parameter Tuning

In the proposed DMLSR, there are two parameters that need to be tuned, i.e., the regularization parameters λ_1 and λ_2. Here, the values of parameters are chosen from the candidate set $\{1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 5e-1, 1e-0\}$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

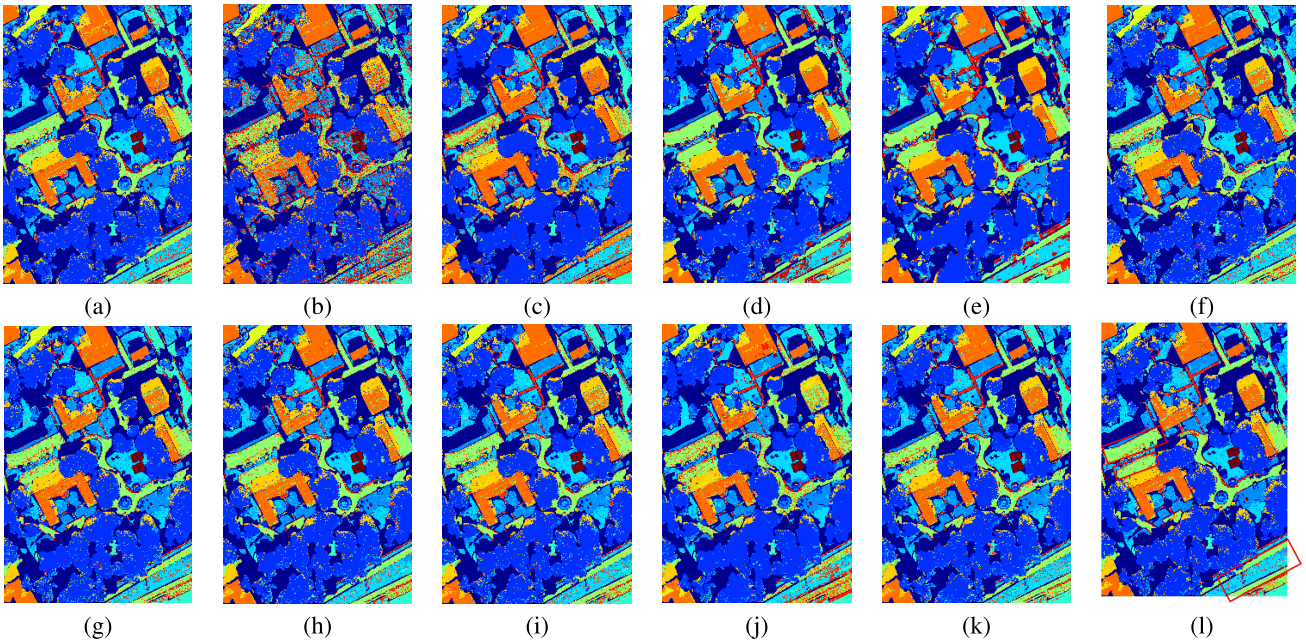


Fig. 9. Visualization and classification maps for the Mississippi data obtained with different methods including (a) SVM (79.95%), (b) SRC (63.76%), (c) CRC (76.38%), (d) OTVCA (81.03%), (e) SSLRCA (80.67%), (f) DLSR (78.86%), (g) ReLSR (78.47%), (h) ICS-DLSR (79.54%), (i) MSRL (79.67%), (j) RLSL (80.21%), (k) RLR (79.52%), and (l) DMLSR (81.17%).

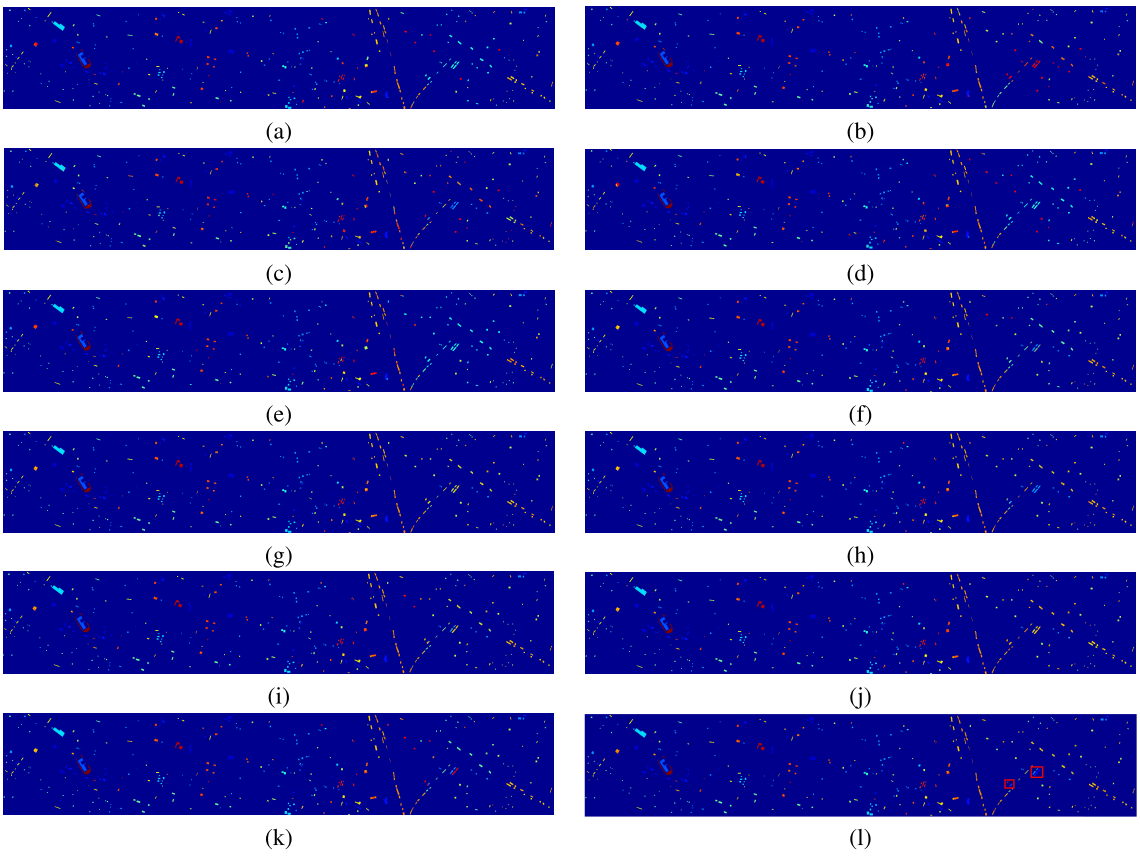


Fig. 10. Visualization and classification maps for the Houston data obtained with different methods including (a) SVM (79.00%), (b) SRC (77.41%), (c) CRC (65.42%), (d) OTVCA (74.03%), (e) SSLRCA (75.39%), (f) DLSR (80.23%), (g) ReLSR (80.23%), (h) ICS-DLSR (80.09%), (i) MSRL (82.69%), (j) RLSL (82.77%), (k) RLR (80.48%), and (l) DMLSR (84.45%).

and the tenfold cross validation is used to select optimal parameters. The relationship between classification accuracy and two parameters is shown in Fig. 6. It is obvious that better classification performance of the three data sets is concentrated in the range of λ_1 and λ_2 between 1e$-$5 and 1e$-$1. According to the maximum OA, the optimal values corresponding to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

parameters are selected. Thus, optimal values of λ_1 and λ_2 for the Indian Pines data are 1e−4 and 1e−1, for the Mississippi data are 1e−2 and 1e−2, and for the Houston data are 1e−5 and 1e−2.

C. Classification Performance on the Synthetic Data Set

Capturing underlying geometry and preserving useful features are important, especially for data with nonlinear manifold structures. A representative three-ring synthetic data set is used, which includes 3-D features, where the first two dimensions are distributed in concentric circles, and the third dimension is amplitude. Fig. 7(a) shows the distribution of the data. In the experiment, the three-ring data contain three classes, and there are 500 training samples and testing samples for each class.

The classification performance of SSLRCA, OTVCA, and several LSR-based methods is provided in Fig. 7. It is obvious from the results that the proposed DMLSR has excellent classification performance while maintaining the manifold structure. Although it can be seen from Fig. 7(h) and (i) that both RLSL and RLR keep the internal structures in the subspace of the projected testing samples, label predictions for some samples are inaccurate. In particular, compared with RLR, the proposed method not only has manifold regularization term learning internal structure but also has data-reconstruction term preserving discrimination information, thereby resulting in better classification performance.

D. Classification Performance on Real-World Data Sets

We further evaluate the classification performance on real-world data sets of the proposed DMLSR with representation-based and LSR-based methods, including SRC, CRC, SSLRA, OTVCA, DLSR, ReLSR, ICS-DLSR, MSRL, RLSL, and RLR. Specifically, in SRC, parameter λ of the three experimental data sets is set to 1e−3. For CRC, the optimal parameter λ corresponding to the Indian Pines data set and the Houston data set is 1e−2 and the Mississippi is 1e−1. For DLSR, ReLSR, and RLR, the optimal parameter λ cross-validated from the candidate set $\{1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 5e-1, 1e-0\}$. For SSLRA, OTVCA, ICS-DLSR, MSRL, and RLSL, the quasi-optimal parameters are obtained according to the optimization strategy [26], [28], [32], [35], [36]. All the algorithms are repeated ten times, and the average classification accuracy and variance are reported.

Tables V–VII reflect the classification performance of the aforementioned classifiers on the three experimental data sets, respectively. For the Indian Pines and Mississippi data, the training samples are randomly selected and average accuracy and standard deviation are reported; for the Houston data, since the training map is fixed, only accuracy is listed. Obviously, the proposed DMLSR has always better classification performance than other algorithms, and the OA value is about 1%∼3% higher than that of others. Specifically, it is evidently superior to representation-based classifiers (i.e., CRC and SRC) even if the fundamental LSR-based classifiers (i.e., DLSR and ReLSR) also have better performance than CRC and SRC. The primary reason is that they only focus on discovering the precise data representation but does not provide the best discrimination. The results indicate that the distinguishable samples are learned via the marginalized constraint and the intraclass compactness graph. Moreover, the proposed DMLSR outperforms MSRL and RLR, due to the fact that transformed samples preserve the important discriminative information.

Classification maps are further shown in Figs. 8–10. In these maps, labeled pixels are displayed as ground truth and unlabeled pixels are displayed as backgrounds. Clearly, the proposed DMLSR obtains less noise and more accurate results in some areas of the classification maps, e.g., the regions of *Corn-no-till* and *grass/pasture* in Fig. 8, the regions of grass ground surface and road materials in Fig. 9, and the regions of stressed grass and soil in Fig. 10.

In order to illustrate the improved performance from a statistical perspective, standardized McNemar's test [40] is

TABLE VIII
STATISTICAL SIGNIFICANCE FROM THE STANDARDIZED MCNEMAR'S TEST ABOUT THE DIFFERENCE BETWEEN THE AFOREMENTIONED METHODS

the Indian Pines	the Mississippi	the Houston
mean Z/significant?		
DMLSR *vs* SVM		
4.70/yes	7.06/yes	14.60/yes
DMLSR *vs* SRC		
13.70/yes	72.10/yes	16.92/yes
DMLSR *vs* CRC		
10.95/yes	24.03/yes	39.19/yes
DMLSR *vs* SSLRCA		
2.06/yes	2.07/yes	21.27/yes
DMLSR *vs* OTVCA		
0.47/no	0.58/no	25.02/yes
DMLSR *vs* DLSR		
10.88/yes	13.33/yes	13.30/yes
DMLSR *vs* MLSR		
8.14/yes	14.49/yes	12.12/yes
DMLSR *vs* ICS-DLSR		
9.52/yes	9.48/yes	13.33/yes
DMLSR *vs* MSRL		
5.70/yes	8.74/yes	6.22/yes
DMLSR *vs* RLSL		
9.96/yes	6.15/yes	5.08/yes
DMLSR *vs* RLR		
9.69/yes	9.86/yes	11.94/yes

TABLE IX
EXECUTION TIME (IN SECONDS) OF DIFFERENT METHODS

Classifier	the Indian Pines data	the Mississippi data	the Houston data
SVM	2.811	5.909	3.993
SRC	383.737	714.503	268.004
CRC	146.609	993.465	459.837
SSLRCA	201.790	241.369	2439.924
OTVCA	248.650	548.775	1660.660
DLSR	1.351	1.706	5.249
ReLSR	4.633	4.512	9.693
ICS-DLSR	2.768	2.670	2.605
MSRL	9.496	15.378	165.046
RLSL	7.220	3.097	7.287
RLR	3.146	7.675	4.220
DMLSR	32.772	14.154	50.918

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

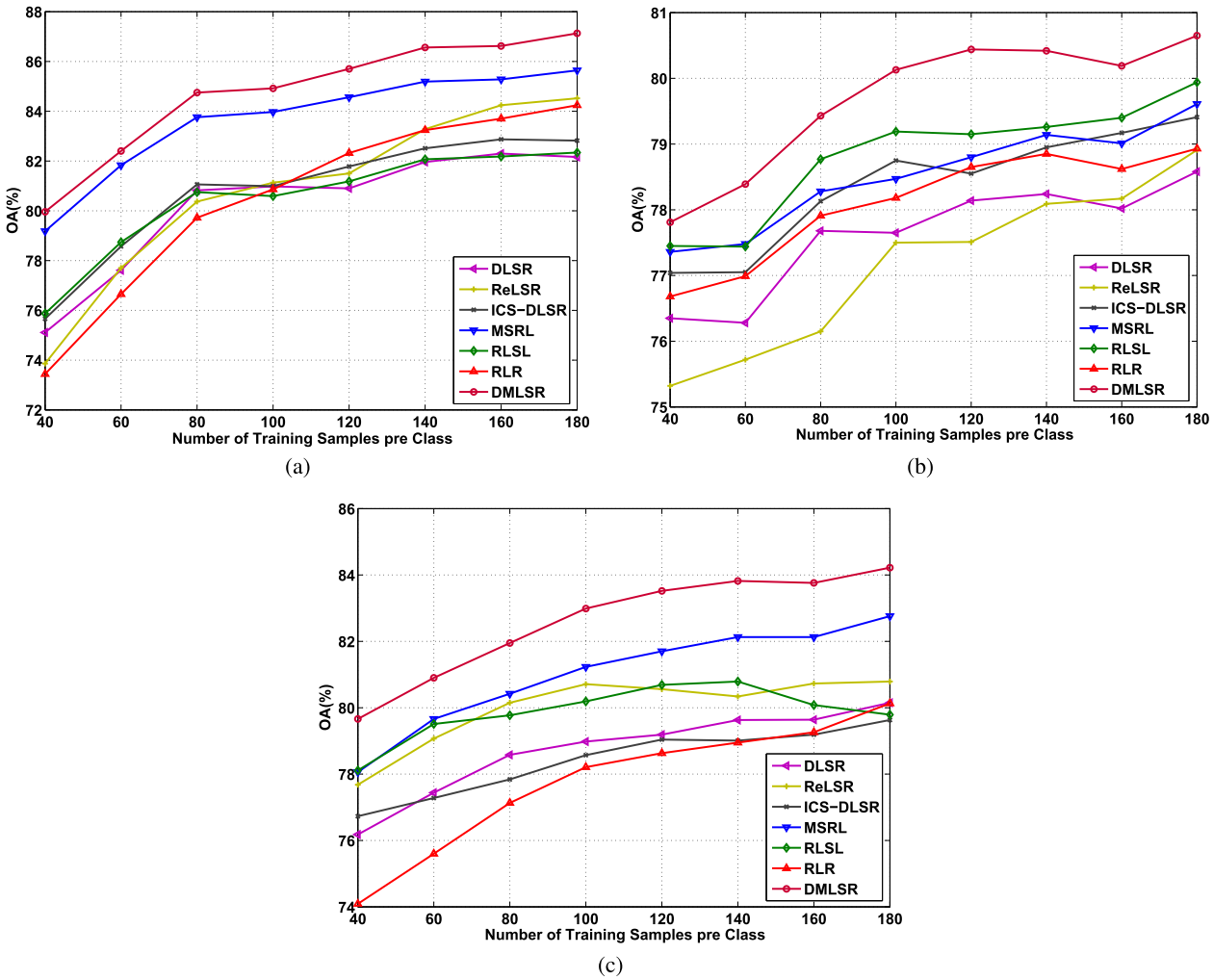


Fig. 11. Overall classification accuracy of LSR-based methods versus different numbers of training samples per class for the three experimental data. (a) Indian Pines data. (b) Mississippi data. (c) Houston data.

performed, and the results are listed in Table VIII. The Z value of McNemar's test is greater than 1.96 and 2.58, indicating that the two compared results are statistically different at 95% and 99% confidence levels. A value of Z greater than 0 indicates that classifier 1 is superior to classifier 2 ($Z > 0$) and vice versa. Comparing the proposed DMLSR with other classifiers, except that the values of OTVCA using the Indian Pines data and Mississippi data are less than 1.96, the remaining comparison results in Table VIII indicate statistical differences above 95% confidence, which further confirms the high discrimination of the proposed method.

Fig. 11 shows the variation of the overall classification accuracy with the number of training samples. In the experiment, the number of training samples in each class varies from 40 to 180, with an interval of 20. It is obvious that OA of all the LSR-based classifiers increases when increasing the training scale, and the proposed DMLSR provides the best performance compared with other classifiers.

Finally, the computational complexity of each classifier is evaluated. All the experiments are carried out using MATLAB on an Intel Core i5-3470 Central processing unit with 8 GB of RAM. The optimization time of the aforementioned methods using the three data set is reported in Table IX, and the number of training and testing samples is consistent with Tables II–IV. It is worth noting that the cost of the proposed DMLSR is significantly less than other representation-based classifiers, although it has a higher complexity than other LSR-based classifiers.

V. Conclusion

In this article, an effective class-label-based regression method for pixelwise hyperspectral data classification was proposed, named DMLSR. A manifold regularization term with intraclass compactness graph was employed to avoid overfitting and obtain near-optimal margin. In addition, a data-reconstruction constraint that allows the projected samples to maintain the discriminative information of the original data is imposed so that the ability to avoid overfitting can be improved. In doing so, the proposed DMLSR obtained more robust subspaces and distinguishable samples than other LSR-based and representation-based classifiers. Extensive experiments on one synthetic data and three real hyperspectral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.

data have demonstrated that the proposed DMLSR is effective in processing manifold-structure data and can significantly improve the classification performance.

REFERENCES

[1] B. Zhang, S. Li, X. Jia, L. Gao, and M. Peng, "Adaptive Markov random field approach for classification of hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 5, pp. 973–977, Sep. 2011.

[2] G. Gao and Y. Gu, "Tensorized principal component alignment: A unified framework for multimodal high-resolution images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 46–61, Jan. 2019.

[3] X. Chen, S. Li, and J. Peng, "Hyperspectral imagery classification with multiple regularized collaborative representations," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 7, pp. 1121–1125, Jul. 2017.

[4] M. Lv, Q. Hou, N. Deng, and L. Jing, "Collaborative discriminative manifold embedding for hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 4, pp. 569–573, Apr. 2017.

[5] X. Jiang, X. Fang, Z. Chen, J. Gao, J. Jiang, and Z. Cai, "Supervised Gaussian process latent variable model for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1760–1764, Oct. 2017.

[6] B. Zhang, "Advancement of hyperspectral image processing and information extraction," *J. Remote Sens.*, vol. 20, pp. 1062–1090, Sep. 2016.

[7] M. Zhang, W. Li, and Q. Du, "Diverse region-based CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2623–2634, Jun. 2018.

[8] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.

[9] R. Hang, Q. Liu, H. Song, and Y. Sun, "Matrix-based discriminant subspace ensemble for hyperspectral image spatial–spectral feature fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 2, pp. 783–794, Feb. 2016.

[10] Y. Zheng, L. Jiao, R. Shang, B. Hou, and X. Zhang, "Local collaborative representation with adaptive dictionary selection for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 10, pp. 1482–1486, Oct. 2016.

[11] Y. Gu, J. Chanussot, X. Jia, and J. A. Benediktsson, "Multiple Kernel learning for hyperspectral image classification: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6547–6565, Nov. 2017.

[12] J. Ma, H. Zhou, J. Zhao, Y. Gao, J. Jiang, and J. Tian, "Robust feature matching for remote sensing image registration via locally linear transforming," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 12, pp. 6469–6481, Dec. 2015.

[13] R. Hang *et al.*, "Robust matrix discriminative analysis for feature extraction from hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 2002–2011, May 2017.

[14] W. Li and Q. Du, "A survey on representation-based classification and detection in hyperspectral remote sensing imagery," *Pattern Recognit. Lett.*, vol. 83, no. 2, pp. 115–123, Nov. 2015.

[15] Q. S. U. Haq, L. Tao, F. Sun, and S. Yang, "A fast and robust sparse approach for hyperspectral data classification using a few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 6, pp. 2287–2302, Jun. 2012.

[16] W. Li, Q. Du, F. Zhang, and W. Hu, "Hyperspectral image classification by fusing collaborative and sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4178–4187, Sep. 2016.

[17] W. Li, Q. Du, and M. Xiong, "Kernel collaborative representation with Tikhonov regularization for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 48–52, Jan. 2015.

[18] K. Li, G. Zhang, X. Li, and J. Xie, "Face recognition based on improved Retinex and sparse representation," *Procedia Eng.*, vol. 15, pp. 2010–2014, Dec. 2011.

[19] L. Zhang, M. Yang, and X. Feng, "Sparse representation or collaborative representation: Which helps face recognition?" in *Proc. Int. Conf. Comput. Vis.*, Nov. 2012, pp. 471–478.

[20] L. Zhang *et al.*, "Kernel sparse representation-based classifier," *IEEE Trans. Signal Process.*, vol. 60, no. 4, pp. 1684–1695, Apr. 2012.

[21] Y. Xu, X. Fang, Q. Zhu, Y. Chen, J. You, and H. Liu, "Modified minimum squared error algorithm for robust classification and face recognition experiments," *Neurocomputing*, vol. 135, pp. 253–261, Jul. 2014.

[22] J. Zou, W. Li, C. Chen, and Q. Du, "Scene classification using local and global features with collaborative representation fusion," *Inf. Sci.*, vol. 348, pp. 209–226, Jun. 2016.

[23] J. Gui, Z. Sun, S. Ji, D. Tao, and T. Tan, "Feature selection based on structured sparsity: A comprehensive study," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 7, pp. 1490–1507, Jul. 2017.

[24] S. Xiang, F. Nie, G. Meng, C. Pan, and C. Zhang, "Discriminative least squares regression for multiclass classification and feature selection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 23, no. 11, pp. 1738–1754, Nov. 2012.

[25] X. Zhang, L. Wang, S. Xiang, and C. Liu, "Retargeted least squares regression algorithm," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 26, no. 9, pp. 2206–2213, Sep. 2015.

[26] J. Wen, Y. Xu, Z. Li, Z. Ma, and Y. Xu, "Inter-class sparsity based discriminative least square regression," *Neural Netw.*, vol. 102, pp. 36–47, Jun. 2018.

[27] X. Fang, Y. Xu, X. Li, Z. Lai, W. K. Wong, and B. Fang, "Regularized label relaxation linear regression," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 4, pp. 1006–1018, Apr. 2018.

[28] Z. Zhang, L. Shao, Y. Xu, L. Liu, and J. Yang, "Marginal representation learning with graph structure self-adaptation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 10, pp. 4645–4659, Oct. 2018.

[29] L. Wang, S. Liu, and C. Pan, "RoDLSR: Robust discriminative least squares regression model for multi-category classification," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Mar. 2017, pp. 2407–2411.

[30] J. Liu, Z. Wu, Z. Xiao, and J. Yang, "Hyperspectral image classification via kernel fully constrained least squares," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 2219–2222.

[31] W. Li and Q. Du, "Laplacian regularized collaborative graph for discriminant analysis of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7066–7076, Dec. 2016.

[32] X. Fang, S. Teng, Z. Lai, Z. He, S. Xie, and W. K. Wong, "Robust latent subspace learning for image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 6, pp. 2502–2515, Jun. 2018.

[33] J. Yang and X. Yuan, "Linearized augmented Lagrangian and alternating direction methods for nuclear norm minimization," *Math. Comput.*, vol. 82, no. 281, pp. 301–329, 2012.

[34] O. Koyejo, S. Acharyya, and J. Ghosh, "Retargeted matrix factorization for collaborative filtering," in *Proc. ACM Conf. Recommender Syst.*, Oct. 2013, pp. 49–56.

[35] B. Rasti, P. Ghamisi, and M. O. Ulfarsson, "Hyperspectral feature extraction using sparse and smooth low-rank analysis," *Remote Sens.*, vol. 11, no. 2, p. 121, 2019. [Online]. Available: https://www.mdpi.com/2072-4292/11/2/121

[36] B. Rasti, M. O. Ulfarsson, and J. R. Sveinsson, "Hyperspectral feature extraction using total variation component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 6976–6985, Dec. 2016.

[37] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[38] T. C. Glenn, A. Zare, and P. D. Gader, "Bayesian Fuzzy Clustering," *IEEE Trans. Fuzzy Syst.*, vol. 23, no. 5, pp. 1545–1561, Oct. 2015.

[39] C. Debes *et al.*, "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.

[40] A. Villa, J. A. Benediktsson, J. Chanussot, and C. Jutten, "Hyperspectral image classification with independent component discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 4865–4876, Dec. 2011.



Yuxiang Zhang (S'19) received the B.S. degree from the Beijing University of Chemical Technology, Beijing, China, in 2017, where he is currently pursuing the M.S. degree under the supervision of Dr. W. Li.

His research interests include hyperspectral imagery processing and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.



Wei Li (S'11–M'13–SM'16) received the B.E. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2007, the M.S. degree in information science and technology from Sun Yat-sen University, Guangzhou, China, in 2009, and the Ph.D. degree in electrical and computer engineering from Mississippi State University, Starkville, MS, USA, in 2012

He spent one year as a Post-Doctoral Researcher at the University of California at Davis, Davis, CA, USA. He is currently with the School of Information and Electronics, Beijing Institute of Technology, Beijing, China. His research interests include hyperspectral image analysis, pattern recognition, and data compression.

Dr. Li was a recipient of the 2015 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society (GRSS) for his service for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS). He is an active Reviewer of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), the IEEE GEOSCIENCE REMOTE SENSING LETTERS (GRSL), and the IEEE JSTARS. He serves as an Associate Editor for the IEEE SIGNAL PROCESSING LETTERS and the IEEE JSTARS. He has served as a Guest Editor for Special Issue of *Journal of Real-Time Image Processing*, *Remote Sensing*, and the IEEE JSTARS.



Heng-Chao Li (S'06–M'08–SM'14) received the B.Sc. and M.Sc. degrees from Southwest Jiaotong University, Chengdu, China, in 2001 and 2004, respectively, and the Ph.D. degree from the Graduate University of Chinese Academy of Sciences, Beijing, China, in 2008, all in information and communication engineering.

From 2013 to 2014, he was a Visiting Scholar with the University of Colorado at Boulder, Boulder, CO, USA. He is currently a Professor with the Sichuan Provincial Key Laboratory of Information Coding and Transmission, Southwest Jiaotong University. His research interests include the statistical analysis of synthetic aperture radar images, remote sensing image processing, and signal processing in communications.

Dr. Li was a recipient of several scholarships or awards, especially including the Special Grade of the financial support from the China Post-Doctoral Science Foundation in 2009 and the New Century Excellent Talents in University from the Ministry of Education of China in 2011. He serves as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS). He has been a Reviewer for several international journals and conferences, such as the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), the IEEE JSTARS, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, the IEEE TRANSACTIONS ON IMAGE PROCESSING, *IET Radar, Sonar and Navigation*, *IET Signal Processing*, *IET Image Processing*, *Pattern Recognition*, the *International Journal of Remote Sensing, Remote Sensing*, and the *Canadian Journal of Remote Sensing*.



Ran Tao (M'00–SM'04) received the B.S. degree from the Electronic Engineering Institute of PLA, Hefei, China, in 1985, and the M.S. and Ph.D. degrees from the Harbin Institute of Technology, Harbin, China, in 1990 and 1993, respectively.

He has been a Senior Visiting Scholar with the University of Michigan, Ann Arbor, MI, USA, in 2001, and the University of Delaware, Newark, DE, USA, in 2016. He is currently a Professor with the School of Information and Electronics, Beijing Institute of Technology, Beijing, China. He has authored three books and more than 100 peer-reviewed journal articles. His research interests include fractional Fourier transform and its applications, theory, and technology for radar and communication systems.

Dr. Tao is a Fellow of the Institute of Engineering and Technology and the Chinese Institute of Electronics. He was a recipient of the National Science Foundation of China for Distinguished Young Scholars in 2006, and a Distinguished Professor of Changjiang Scholars Program in 2009. He has been a Chief-Professor of the Creative Research Groups with the National Natural Science Foundation of China since 2014 and was a Chief-Professor of the Program for Changjiang Scholars and Innovative Research Team in University from 2010 to 2012. He is currently the Vice-Chair of the IEEE China Council. He is also the Vice-Chair of the International Union of Radio Science (URSI) China Council and a member of the Wireless Communication and Signal Processing Commission, URSI. He was a recipient of the First Prize of Science and Technology Progress, in 2006 and 2007, respectively, and the First Prize of Natural Science in 2013, both awarded by the Ministry of Education.



Qian Du (S'98–M'00–SM'05–F'18) received the Ph.D. degree in electrical engineering from the University of Maryland, Baltimore, MD, USA, in 2000.

She is currently the Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a Fellow of the SPIE-International Society for Optics and Photonics. She was a recipient of the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society. She was the Co-Chair of the Data Fusion Technical Committee, IEEE Geoscience and Remote Sensing Society, from 2009 to 2013, and the Chair of the Remote Sensing and Mapping Technical Committee, International Association for Pattern Recognition, from 2010 to 2014. She is the General Chair of the fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing, Shanghai, in 2012. She has served as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the *Journal of Applied Remote Sensing*, and the IEEE SIGNAL PROCESSING LETTERS. Since 2016, she has been the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:30:51 UTC from IEEE Xplore. Restrictions apply.